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PROJECT REPORT • SEPTEMBER 2026 • BY PAWAN KUMAR VAISHNAV

Bike Sharing Demand Analysis Using Excel: Methods, Findings and Recommendations



Bike-sharing operators record every rental, hour after hour, but the value only appears when those records are cleaned, combined and turned into answers. This report shows how Microsoft Excel was used to do that for 1,000 hourly rental records, and what the analysis says about when, why and how riders use the service.

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Introduction and Objective

Why bike-sharing data needs careful analysis

Background

A bike-sharing system logs every trip automatically. Operators must decide how many bikes to keep at different hours, when to service them, how far bad weather cuts demand and which riders to target with promotions.

Raw records, however, arrive in separate tables, use numeric codes instead of readable labels and may contain gaps or duplicates. Excel handles cleaning, formulas, pivot tables, charts and automation in one tool, and its results can be shared as a dashboard.

Data used

1,000 hourly records from 1 January to 14 February 2011, merged from three datasets in the workbook DIGI-PROJECT_1_2.xlsx.

Six project objectives

1. Import three datasets and merge them into one clean table.
2. Detect and handle missing, duplicate or inconsistent data.
3. Create readable categories with conditional formulas.
4. Summarise demand by hour, day, weather, humidity and user type with pivot tables.
5. Present results in an interactive dashboard with slicers.
6. Automate cleaning and reporting with VBA macros.

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Data and Datasets

Three source sheets, one final table

DATA_SET_1 | 610 rows

Date, season, year, month, hour, holiday flag, weekday, weather situation and temperature of each hour.

DATA_SET_2 | 610 rows

The same 610 hours with assumed temperature, humidity, wind speed and casual, registered and total rentals.

DATA_SET_3 | 390 rows

All fields, continuing from S.No. 611. Appended below sets 1 and 2 to give 1,000 records.

Data coverage

- The period covers 45 dates, all in the same season (spring code) and year (2011).
- Only 17 dates have all 24 hourly records, 20 dates have 23 records and 8 dates have fewer (between 8 and 22).
- Temperature, humidity and wind are stored as scaled values (0 to about 0.58). The wind column is labelled km/h, but the values are not real speeds, so wind and humidity categories are relative groupings.
- Main measure: Total Rentals, equal to Casual Users plus Registered Users in every row.

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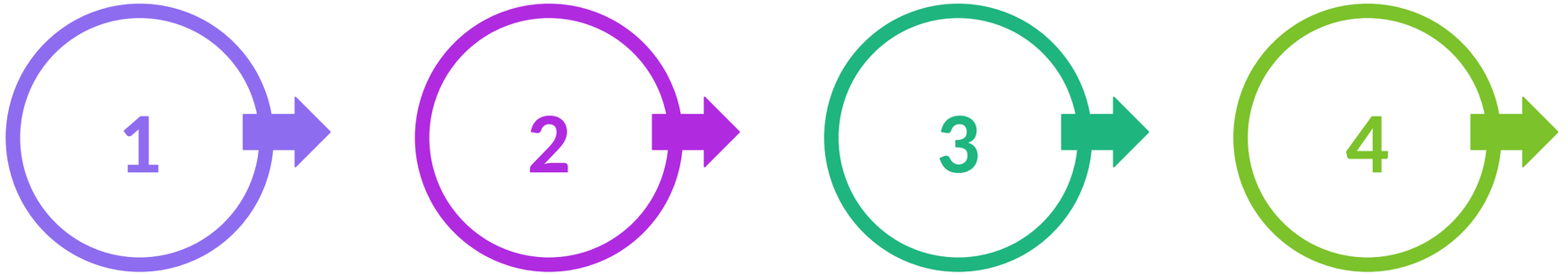
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Data Preparation

Merging, quality checks and derived columns



Merge

DATA_SET_1 and 2 joined on S.No., set 3 appended: FINAL DATA SET of 1,000 hourly records.

Check

No duplicates or empty cells. Casual + Registered = Total in all 1,000 rows.

Derive

IF, WEEKDAY, TEXT and HOUR formulas add day period, day type, weather, season and day of week.

Categorise

Temperature, humidity and wind categories and a holiday flag support the pivots.

New column	Rule	Main function
Day Period	6 to 11 Morning, 12 to 16 Afternoon, 17 to 20 Evening, else Night	IF, AND, HOUR
Day Type	Saturday and Sunday are Weekend, other days Weekday	IF, WEEKDAY
Weather Label	1 Clear, 2 Mist/Cloudy, 3 Light Snow/Rain, 4 Heavy Rain/Snow	IF
Day of Week	Name of the weekday, for example Monday	TEXT
Temperature Category	Below 0.3 Cold, below 0.6 Moderate, else Hot	IF

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Analysis and Dashboard

How the workbook turns data into answers

Pivot tables

Sheets for hour and day, weather and wind, humidity, and holiday and day type, each with a pivot chart.

1

KPI sheet

Headline figures are read from the pivots with GETPIVOTDATA, so they update when the pivots refresh.

Dashboard

Bar, line, combination, doughnut and pie charts on one page, all filtered together.

2

Slicers

Six slicers: holiday flag, day type, weather, day of week, humidity category and month.

Why average rentals per hour?

Clear weather has 606 hours and light rain only 84, so totals mostly reflect group size. Averages compare like with like.

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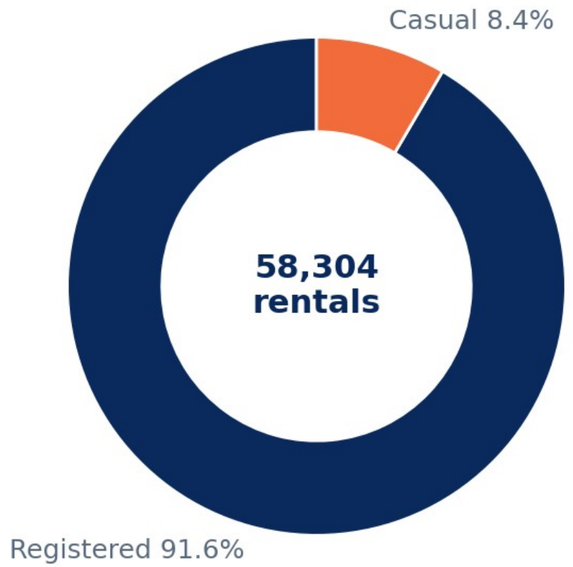
Headline figures

58,304
Total rentals

91.6%
Registered users

58.3
Average per hour

249
Highest single hour



Registered members produce more than nine in every ten rentals (53,383 against 4,921 casual), so the service is a commuter and member service with a smaller leisure segment. The average day has about 1,296 rentals, and the highest single hour was 8:00 AM on 20 January 2011 with 249 rentals. Busiest day: 25 January (1,985 rentals).

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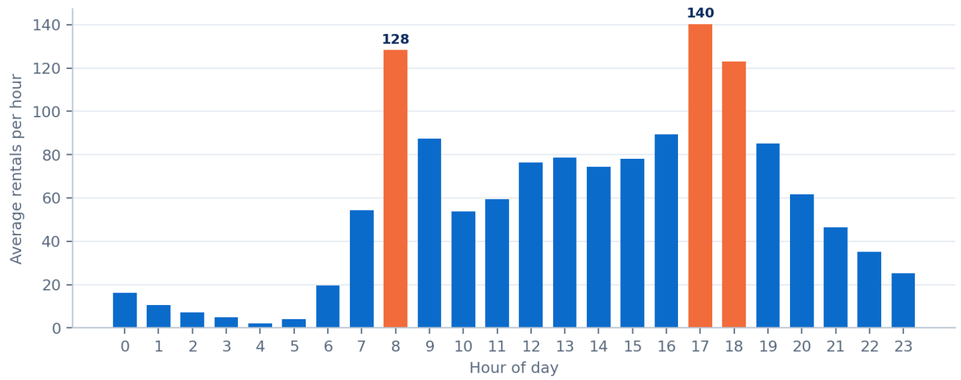
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Key Findings

Demand by time of day



Evening peak at 5 PM: 140 rentals per hour, the highest of the day.

Midday demand holds at about 75 to 90 rentals per hour. Afternoon totals 17,139 rentals and morning 16,968. This double peak is typical of commuting, so fleet plans should be built around 8 AM and 5 PM.

8 AM

Morning peak

Demand is near zero overnight, rises sharply at 7 AM and peaks at 8 AM with 128 rentals per hour as commuters start their day.

5 PM

Evening peak

The highest point is 5 PM with 140 rentals per hour, followed by 123 at 6 PM. Evening is the busiest period with 17,769 rentals.

2 AM

Night low

Between 2 AM and 5 AM only 2 to 7 rentals per hour occur. Night hours are 36% of records but 11% of rentals, the best window for servicing.

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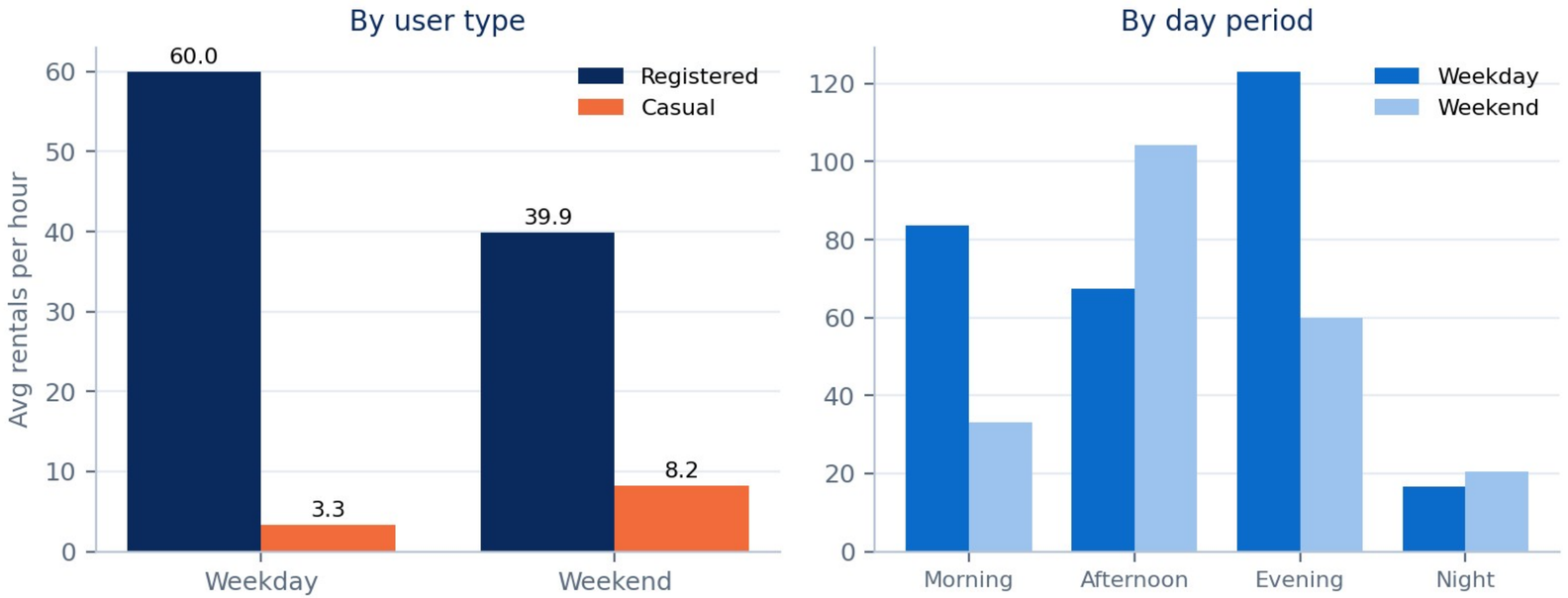
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Weekdays, weekends and user types



Weekdays average 63.3 rentals per hour against 48.1 at weekends (about 24% lower). Thursday is strongest (67.7) and Saturday weakest (46.7). Registered users fall from 60.0 to 39.9 per hour at weekends, while casual users rise from 3.3 to 8.2.

The shape of the day changes too: weekday evenings reach 123.1 per hour and mornings 83.7, but weekend afternoons lead with 104.3 and mornings are weak at 33.0. Casual users peak at about 3 PM and make 58.6% of their rentals between 12 PM and 5 PM. The single holiday (17 January) averaged 41.7 per hour, but rests on only 24 records.

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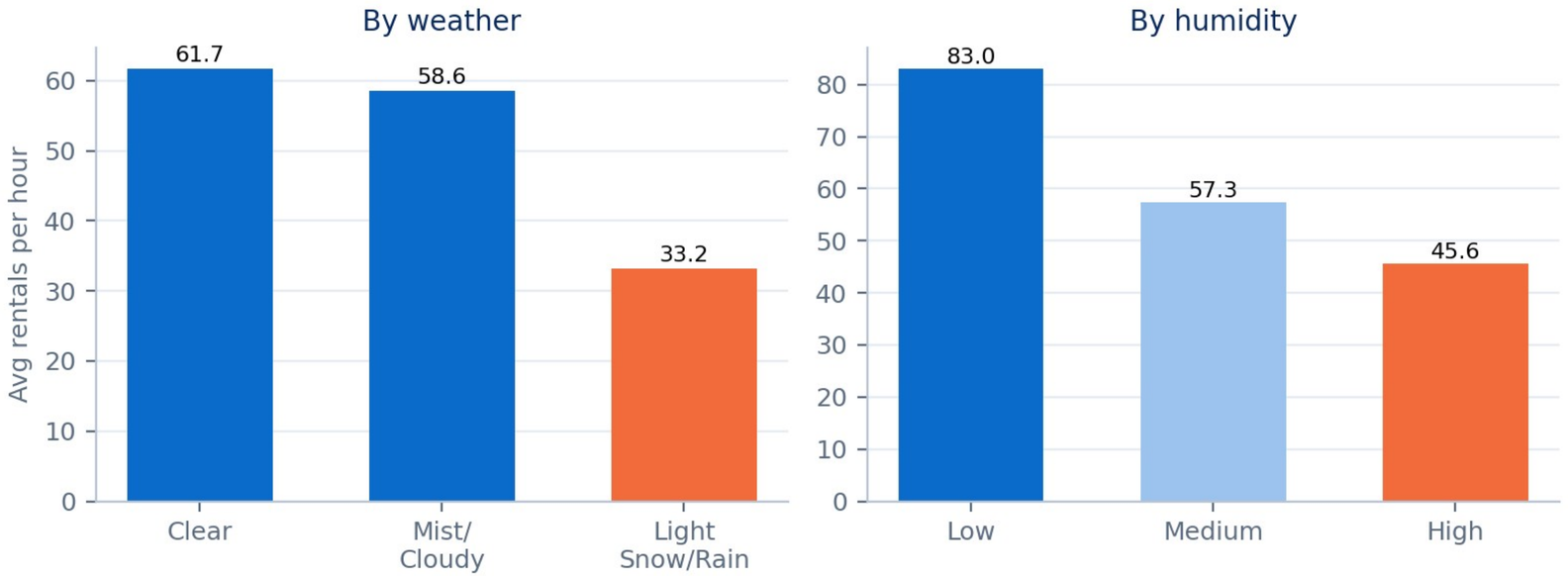
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Key Findings

Weather, humidity and temperature



Clear weather (606 hours) averages 61.7 rentals per hour and mist or cloud 58.6, but light snow or rain drops the average to 33.2, about 46% lower. Low humidity averages 83.0, medium 57.3 and high 45.6.

Moderate-temperature hours average 85.7, about 59% more than cold hours (53.9); no hour is in the hot category. Windy hours average 69.0 against 55.0 for calm hours, but this is likely because windier hours fall in the afternoon, so wind is not a reason to stay away. Weather, humidity, temperature and time of day are related, so these differences show association, not cause.

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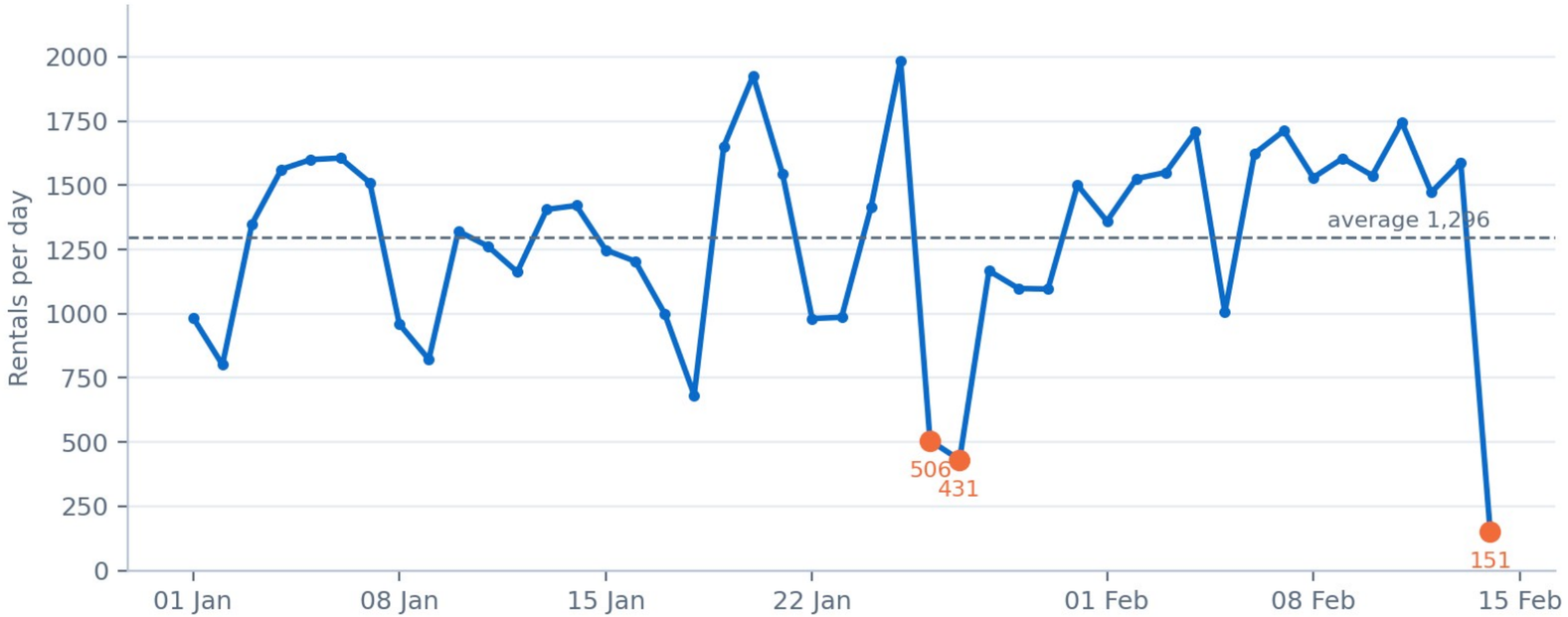
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Key Findings

Unusual days



Three dates are far below the daily average of 1,296. 26 January recorded 506 rentals, with light snow or rain in 13 of its 16 recorded hours, so bad weather is the likely cause. 27 January (431) and 14 February (151) contain only 8 hourly records each, so their totals are low mainly because most of the day is missing.

The busiest day, 25 January (1,985 rentals), had 23 recorded hours and mostly mist or cloud, showing that mild weather on a weekday can produce very high demand. Comparing days by total is only fair for dates with a complete set of hours.

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Automation with VBA

Macro module: BikeSharing_Macros.bas

Macro	What it does
RunFullReport	Master macro. Runs the four steps below in order and shows a message when finished.
CleanNewData	Trims text, removes empty and duplicate S.No. rows and highlights missing numeric values in yellow.
RefreshDashboard	Refreshes every pivot table, chart and formula so the dashboard shows the new data.
BuildAnomalyReport	Totals rentals per date, calculates z-scores and writes an Anomaly_Report sheet flagging unusually low or high days and partial days.
ForecastNext7Days	Fits a linear trend to full 24-hour days and forecasts the next seven dates.

Custom functions: DayPeriod, WeatherName and DayType can be used directly in cells and mirror the IF formulas in the derived columns.

To use: press Alt+F11, choose File, Import File, select the .bas file, save the workbook as .xlsm and run RunFullReport with Alt+F8. The forecast is deliberately simple (17 full days only); Excel's Forecast Sheet can be used as a cross-check.

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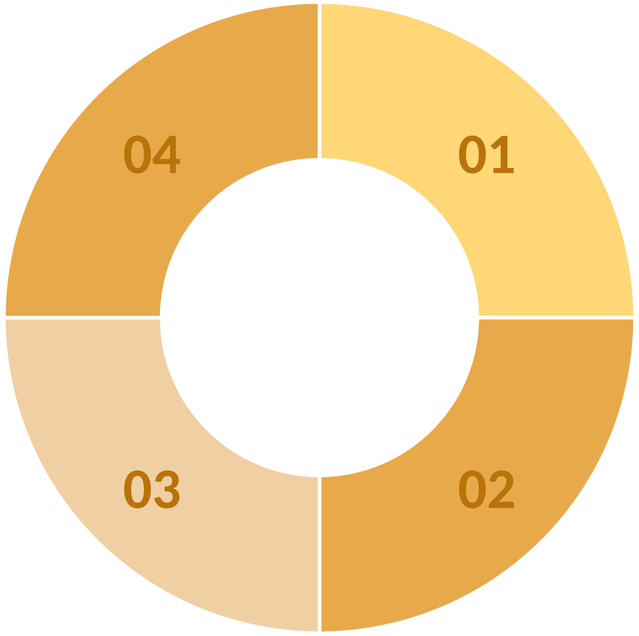
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Each action follows directly from the findings



Match fleet to the peaks

Stock and rebalance bikes before 8 AM (128 per hour) and 5 PM (140). Use 2 AM to 5 AM, when only 2 to 7 rentals occur, for servicing.

Plan weekends separately

Weekend demand is lower (48.1 vs 63.3 per hour) and shifts to the afternoon (104.3), so weekday and weekend fleet plans should differ.

Adjust to the weather

Light rain or snow cuts demand by 46% and high humidity lowers it too. Scale deployments to the forecast.

Serve both user groups

Casual users (8.4%) peak at weekends and 12 PM to 5 PM: aim membership offers there. Registered members (91.6%) need reliable weekday service.

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What the analysis can and cannot say

Limitations

- Only about six weeks of winter and spring data, so seasons and growth cannot be measured.
- 28 of 45 dates have fewer than 24 hourly records.
- Holiday (24 records) and heavy rain (1 record) rest on very few rows.
- Temperature, humidity and wind are scaled values.
- Group differences show association, not cause.
- The VBA forecast is a linear estimate on 17 full days.

Conclusion

Using only Excel, three datasets were merged, cleaned, enriched, analysed, visualised and automated. Demand is shaped first by time of day and type of day, and second by weather.

Registered commuters form the core, peaking at 8 AM and 5 PM on weekdays, while casual riders appear on weekend afternoons. Light rain, snow and high humidity reduce rentals, and 2 AM to 5 AM is the best maintenance window.

Next step: collect a full year of complete hourly data to enable seasonal forecasting.